

Diagnostic Goodness-of-Fit Tests for Incidence and Latency Components in Mixture Cure Models with Application to Pediatric Acute Myeloid Leukemia

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Introduction

- In survival analysis, it is possible that some patients will never experience the event of interest even after long-term follow-up.
- Mixture cure models address this issue by representing the population as a combination of individuals who have been cured and individuals who are still susceptible.
- The model contains two components: 1) Incidence component: Models cure probability. 2) Latency component: Models survival among susceptible patients.
- However, model misspecification may lead to biased inference about the cure fraction and survival behavior.
- Therefore, goodness-of-fit (GOF) tests are needed to assess whether the assumed parametric models are appropriate.

Study Data and Objectives

- Data come from the Pediatric Acute Myeloid Leukemia (AML) study and focus on patients in the MLL cytogenetic subgroup.
- Fit a mixture cure model to the study data.
- Evaluate the goodness-of-fit of the incidence and latency components from the fitted mixture cure models.
- Assess whether the assumed parametric models provide a reasonable fit to the Pediatric AML MLL subgroup data.

Figure 1: Kaplan-Meier Survival Curve: Pediatric-AML MLL Subgroup
The yellow shaded region represents the 95% confidence interval for the survival estimate.

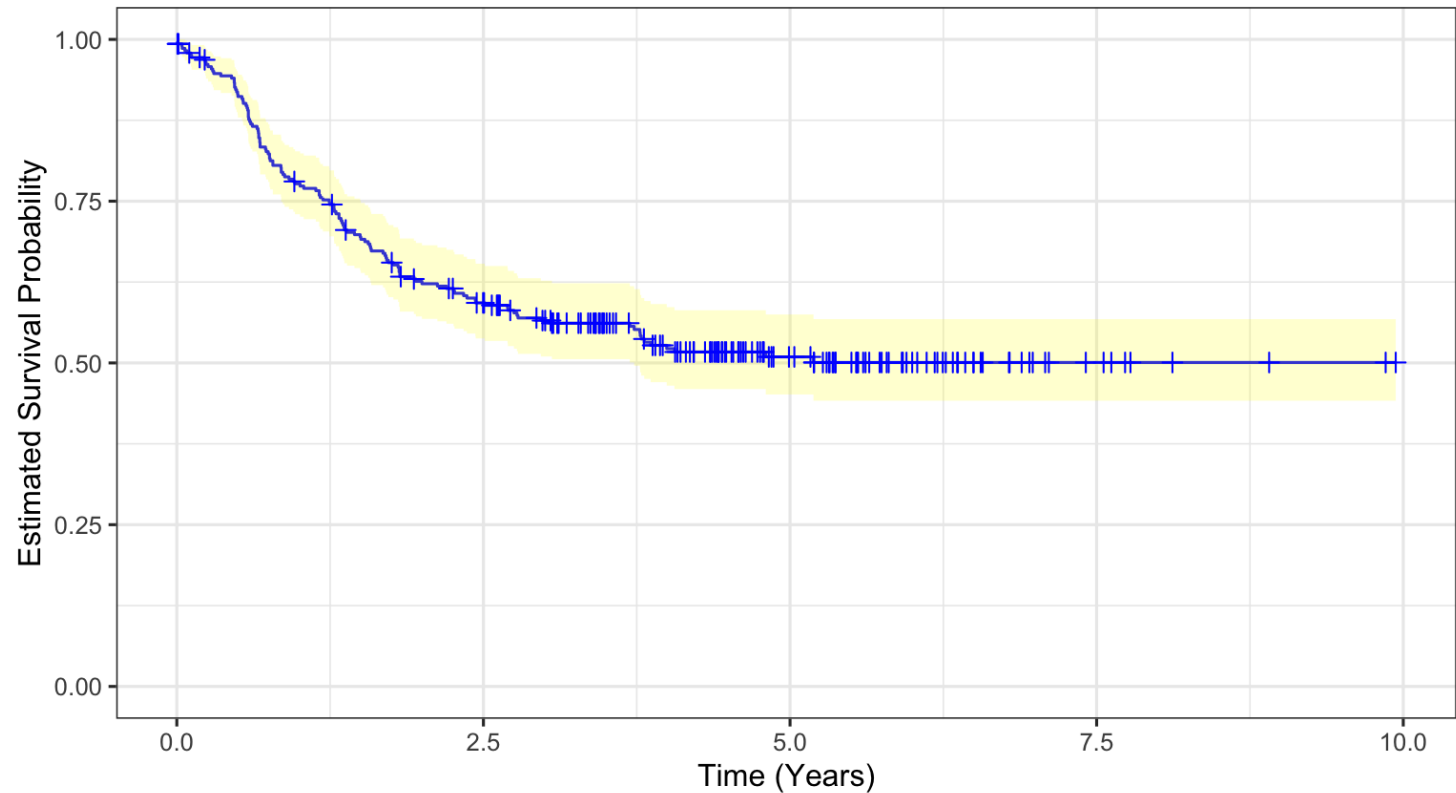


Figure 1 shows the Kaplan-Meier estimate of overall survival for this subgroup. The survival curve decreases rapidly for the first few years and then reaching a plateau after about five years. This plateau suggests that a fraction of patients may not experience the event during long-term follow-up, which motivates the use of a mixture cure model.

Simulation results

Table 1: Simulation results for the latency GOF test under a long plateau scenario ($\varphi = 0.75$). Rejection rates at the 5% and 10% significance levels are reported. The results are based on 1000 trials and 500 bootstrap samples.

True Distribution	Fitted Distribution	Number of Sample Size	Rejection Rate (5%)	Rejection Rate (10%)
Weibull	Weibull	150	4.7	10.0
	Gompertz	150	43.1	52.4
	Lognormal	150	81.7	87.2
	Uniform	150	99.6	99.7
	Weibull	500	5.2	8.8
	Gompertz	500	94.4	96.8
	Lognormal	500	99.9	100.0
	Uniform	500	100.0	100.0

Table 2: Simulation results for the incidence GOF test, where the cured probability = $1 / (1 + \exp(\theta_0 + \theta_1 x_1 + \theta_2 x_2))$ and the time $\sim$ Weibull (shape = $a_x = (1 + x)^{c_T}$, scale = $\exp(-(1+x) / (2 * a_x))$) The significance level is $\alpha = 0.05$. The results are based on 500 trials and 500 bootstrap samples. $x_1 \sim \text{unif}[-1,1]$ and $x_2 = x_1^2$.

c_T	θ_0	θ_1	θ_2	N = 50	N = 100	N = 200
0	1	1	0	4.2	6.4	5.8
			1	9.6	12.0	18.8
			2	18.2	39.8	73.0
			4	63.4	93.8	100.0

N denotes the sample size, c_T controls the shape of the latency distribution, $ch = 1.5$ is the censor mean, and θ_2 represents the degree of misspecification in the incidence model. The null hypothesis corresponds to $\theta_2 = 0$, where the model is correctly specified, and θ_0 and θ_1 control the baseline and linear covariate effect.

Application to Study Data

Table 3: GOF test results for the mixture cure model applied to the Pediatric AML MLL subgroup data. The table reports results for both the incidence and latency components.

Component	Model/Dis.	Test Statistic	P-value	AIC
Incidence	Logistic	0.789	0.066	-
	Weibull	0.155	0.214	721.185
Latency	Gompertz	0.212	0.054	722.928
	Lognormal	2.933	0.002	747.554
	Uniform	12.111	< 0.001	786.450

Mixture Cure Model

The mixture cure model is defined as:
 $S(t | X) = P(T > t | X) = p(X) + \{1 - p(X)\}S_1(t | X)$
Incidence Component: Latency Component:
 $\text{logit}(p(X)) = \beta_0 + X^T \beta \quad S_1(t) = \exp\{- (\lambda t)^\rho\}$

Where $p(X)$ is the cure probability and $S_1(t | X)$ is the survival function for the susceptible patients. The latency component follows a Weibull distribution with shape parameter $\rho > 0$ and scale parameter $\lambda > 0$.

Goodness-of-Fit tests

Component	Method Used	Purpose
Incidence Component	Müller and Van Keilegom (2019)	Tests whether the logistic model correctly specifies the cure probability
Latency Component	Geerdens et al. (2020)	Tests whether the assumed parametric survival distribution fits the susceptible patients
Inference	Bootstrap Procedure	To compute bootstrap p-values for the tests

Conclusions

- Simulation results show that the GOF tests maintain reasonable type I error and gain higher power when the model is misspecified and the sample size increases.
- The results suggest that the logistic incidence model and the Weibull latency model fit reasonably well for this dataset.
- These GOF tests provide a practical way to check mixture cure model assumptions in real data applications.

Acknowledgements

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References

Müller, J., & Van Keilegom, I. (2019). *Goodness-of-fit tests for the incidence component of mixture cure models*.
Geerdens, C., Acar, E. F., & Van Keilegom, I. (2020). *Goodness-of-fit tests for the latency component of mixture cure models*.
Xu, J., & Peng, Y. (2014). *Nonparametric goodness-of-fit tests for mixture cure models*.